

PROJECT SUTRA — STRATEGIC TWIN PITCH SCRIPT

HARIKA: VISION, PROBLEM & SOLUTION OVERVIEW • NIKHIL: DEEP DIVES & FIELD APPLICATIONS

PRESENTERS: HARIKA • NIKHIL
TOTAL TIME: ~2 MIN 15 SEC

🎯 **ROLE DIVISION:** Harika anchors the room with the opening hook, disaster problem audit, and solution value proposition. Nikhil steps in as Chief Swarm Architect to lead the deep subsystem engineering breakdowns and real-world field deployment.

SLIDE 01 Introduction: Project SUTRA

00:00 - 00:30 (30s)

SPEAKER: HARIKA (MISSION & OPERATIONS LEAD) [STAGE: CENTER STAGE • COMPELLING OPENING HOOK]

🔥 **HOOK:** "When a disaster strikes, the first 72 hours decide who lives. But the second drones enter thick forests or mountain ravines, they lose signal and crash."

"Respected jury, we are **Team OFFGRID**. I am Harika, and with me is Nikhil, our Chief Swarm Architect. We built **Project SUTRA** — an autonomous drone swarm that finds survivors in disaster zones with **zero GPS and zero internet**."

"Instead of single manual drones, SUTRA deploys an intelligent swarm backed by 3 breakthroughs:"

PILLAR 1 Autonomous Swarm Flight: Drones dodge obstacles and coordinate together 50 times a second without ground towers. [Slide: 50Hz GNC]

PILLAR 2 AI-Powered Comms: Live video stays connected through mountain walls without blacking out. [Slide: -5dB JSCC]

PILLAR 3 3D Elevation Vision: Locks onto survivors on steep slopes with centimeter accuracy. [Slide: 3.59cm DEM]

CONTINUES WITH HARIKA → "Let me show you why today's search and rescue missions fail in the field."

SLIDE 02 The Problem: 4 Field Failure Voids

00:30 - 01:15 (45s)

SPEAKER: HARIKA [STAGE: URGENT, GROUNDED, EMPIRICAL AUDIT]

⚠️ **HOOK:** "When we audited actual NDRF disaster reports across Wayanad, Sikkim, and Chamoli, we found 4 fatal breakdowns that cause current missions to fail:"

VOID 1 Tree Canopies Block GPS: Drones lose satellite lock, drift blindly into branches, and crash. *(In Wayanad, 70% of canopy sorties failed).*

VOID 2 Mountain Ridges Cut Video: The moment a drone dips behind a ridge, standard video glitches and cuts to total black.

VOID 3 Flat-Map Errors on Slopes: Standard drone cameras assume the earth is flat. On steep hillsides, coordinates miss by **15 to 30 meters** — routing teams to empty ravines.

VOID 4 Human Pilot Overload: Flying 5 drones takes **15 to 25 crew members**, taking hours to set up and costing ₹12.5 Lakhs per sortie.

CONTINUES WITH HARIKA → "Here is how SUTRA's 4 core moats solve every single breakdown."

SPEAKER: HARIKA [STAGE: HIGH-ENERGY SOLUTION PRESENTATION]

HOOK: "SUTRA directly answers all 4 problem voids with 4 focused engineering moats:"

- MOAT 1Self-Flying Swarms: Drones calculate collision-free flight paths onboard at 50Hz — zero ground towers, zero GPS reliance. [Slide: SUTRA-FSD & 3D ORCA]
- MOAT 2AI-Guided Neural Video: Even in deep ravines where standard signals die, our neural link transmits clear video through solid mountain terrain. [Slide: Deep JSCC | -5dB]
- MOAT 33D Mountain Vision: Onboard AI maps terrain height in real time, locating survivors within 3.5 centimeters in under 15 milliseconds. [Slide: 3D DEM Jetson Orin]
- MOAT 4One-Person Command: A single tactical tablet lets 1 operator command the entire swarm for just ₹95,000 per sortie. [Slide: Pegasus 3D GCS]

"The bottom line: SUTRA scans a full square mile in 10 minutes — 4 times faster than human rescue crews, saving lives before time runs out."

Metric	Traditional SAR	Single Drone	SUTRA SWARM
Area Coverage / mi ²	2-3 hours	45-60 min	10-18 min (4x Faster)
Detection Rate	65-70%	55-65%	78-85% First Pass
Personnel Required	15-25 people	2-3 pilots	1-2 Operators
Sortie Cost	₹12.5 Lakhs	₹5.0 Lakhs	₹95,000

🔥 POWER HANDOFF TO NIKHIL → "Now, our Chief Swarm Architect, Nikhil, will break down the deep engineering behind each subsystem and our real-world field deployment."

STAGE TAKEOVER: NIKHIL (CHIEF SWARM ARCHITECT)

[DEEP SUBSYSTEM DEEP DIVES, CODE ARCHITECTURE & FIELD CONOPS]

- Slide 04 (Subsystem A): 50Hz PX4 MicroXRCE Offboard + 3D ORCA Velocity Obstacles
- Slide 05 (Subsystem B): Deep JSCC Autoencoders & Mesh Resiliency at -5dB SNR
- Slide 06 (Subsystem C): YOLOv8-Pose + WGS84 3D DEM Geolocation Raycasting on Jetson Orin
- Slide 07 (Subsystem D): WebGPU 3D Digital Twin & Offline ATAK Field Command
- Slide 08 (Field CONOPS & Q&A): Wayanad Simulation Audit, Hardware BOM, and Jury Defense